

Submission Format

AHC Video Anomaly Detection: everything you need to submit.

What you do

Sign in to the arena site with the Google account we registered for you. You can only see and submit for yourself.

1. **Benchmark** tab → **Submission format** → **download manifest.json**. The list of videos you must answer for, and each one's level.
2. Same panel → **download starter template**. Every video is pre-filled with empty answers. Start from this rather than typing JSON by hand.
3. Run your model and fill in the template.
4. **Submit predictions** → pick your file → **Submit and score**. You get your score straight away.
5. Repeat as needed. **Each upload re-scores everything you've answered so far, and replaces your previous score**. There is no "best of", so a later run stands even if it scores lower.
6. Fill in the **Final submission** section at the bottom of the page: code repo and architecture write-up. Required.

Every upload is kept and listed under **My Submissions**, with the one that currently counts marked. The live leaderboard is on the big screen.

If a file has a problem you get a table saying exactly which video and which field. **A rejected file does not use up a run**. Maximum file size 5 MB.

You can submit in pieces

A file only updates the videos it mentions. Your other answers stay exactly as they were.

Note: E001, E002, ... are video ids (video file names)

	You upload	Your answers afterwards
Run 1	E001, E002	E001 ✓ · E002 ✓
Run 2	E002, E003, E004	E001 kept · E002 replaced · E003 ✓ · E004 ✓

- Re-answering a video **replaces** your old answer. Latest upload wins.
- Leaving a video out **keeps** your previous answer. It is not cleared.
- A video you never answer is scored as **normal**.

So you can work through the pack as you go. Your page shows *"Videos answered 12 of 28"*. Each upload costs one run regardless of how many videos it covers, so batch sensibly.

Your score is always the score of your current answer sheet. Every upload recomputes it across all 28 videos and overwrites your previous score. Uploading a file that makes things worse will lower your standing. There is no fallback to an earlier, better run.

The file

```
{
  "schema_version": "1.0",
  "submission_id": "my-team-run-01",
  "model_name": "my-near-real-time-vad",
  "run_metadata": {
    "total_wall_time_ms": 1476000,
    "hardware": "1x RTX 4090"
  },
  "predictions": [
    {
      "video_id": "E001",
      "events": [],
      "runtime_metadata": {
        "frames_processed": 90,
        "chunks_processed": 1,
        "end_to_end_internal_time_ms": 4820,
        "model_runtimes": []
      }
    },
    {
      "video_id": "E009",
      "events": [
        {
          "class_name": "fire",
          "start_time_sec": null,
          "end_time_sec": null,
          "explanation": "Thick smoke rises from a burning structure."
        }
      ],
      "runtime_metadata": {
        "frames_processed": 30,
        "chunks_processed": 1,
        "end_to_end_internal_time_ms": 2100,
        "model_runtimes": []
      }
    },
    {
      "video_id": "E021",
      "events": [
        { "class_name": "traffic_accident", "start_time_sec": 20, "end_time_sec": 40 },
        { "class_name": "traffic_accident", "start_time_sec": 60, "end_time_sec": 80 }
      ],
      "runtime_metadata": {
        "frames_processed": 600,
        "chunks_processed": 24,
        "end_to_end_internal_time_ms": 91000,
        "model_runtimes": [
```

```

    {
      "model_name": "vision-language-model",
      "call_count": 24,
      "total_time_ms": 81900,
      "average_time_ms": 3412.5,
      "p50_time_ms": 3310.0,
      "p95_time_ms": 4028.0,
      "max_time_ms": 4605.0
    }
  ]
}
]
}
}

```

E001 is called normal. **E009** is a Level-1 event with no timestamps. **E021** is a Level-2 video with two separate accidents.

Field rules

The only strictly required top-level field is **predictions**. The rest are recommended.

Per video, all three of **video_id**, **events** and **runtime_metadata** are required. **video_id** must match the manifest exactly and appear once only.

Per event

Field	Rule
class_name	One of the 11 classes below.
start_time_sec	null at Level 1. Required and ≥ 0 at Levels 2 to 3
end_time_sec	null at Level 1. Must be greater than start

explanation	Optional, 20 to 500 characters. Bonus only. Omitting it never costs you
-------------	---

runtime_metadata, required on every video

Field	Rule

If a `model_runtimes` entry gives `total_time_ms`, `call_count` and `average_time_ms`, the average must match $\text{total} \div \text{calls}$ within 2%. Any `call_times_ms` must have exactly `call_count` entries.

Time from the start of processing that video until its event list is ready. **Include** decoding, preprocessing, inference and postprocessing. **Exclude** model loading and downloads.

Unknown extra fields (`confidence`, `bbox`, debug keys) are ignored, not rejected.

The 11 classes

traffic_accident	traffic_congestion
stalled_or_broken_down_vehicle	vehicle_blocking_traffic
fire	smoke
waterlogging_or_flood	wrong_way_driving
road_spill_or_debris	fighting_or_violence
loitering_or_suspicious_presence	normal

The three levels are three different tasks

Level	Rows carry	What you output
1	No timestamps	One label for the whole clip
2	Timestamped intervals	Every event, with start and end times
3	Timestamped intervals	Same, on longer and harder footage

At **Level 1**, repeating a class on one video earns nothing extra.

At **Levels 2 and 3**, an event counts only when the class is right **and** temporal overlap reaches **0.5**, where overlap is $\text{intersection} \div \text{union}$ of the two intervals.

In practice: **if your interval sits inside the real event it must cover at least half of it. If it swallows the real event it must be no more than twice as long.** Several partial fragments for one real event don't help. At most one can match, and the rest count against you.

Scoring

Level 1 is pooled across all Level-1 videos: half anomaly-versus-normal accuracy, half class accuracy.

Levels 2 and 3 are scored per video, then averaged.

- Ground truth is **normal** → you predict nothing = **1**, you predict anything = **0**.
- Ground truth has events → a weighted mix of *did you alert*, *matched events*, and *how well your timings line up*. Timing weighs more at Level 3.
- All three are always shown separately, so a bonus can never disguise weak accuracy.

Latency bonus: your total reported processing time ÷ total video duration.

Final submission

At the bottom of the Benchmark tab. Required, and separate from your score.

- **Code repository:** a URL a host can actually open.
- **Architecture write-up:** a link, or a PDF/HTML upload up to 25 MB. Show how your pipeline fits together: which models, in what order, what runs per frame versus per clip. A diagram beats prose.
- **Notes:** optional. Limitations, how to run it, what you'd do with more time.

Note: Please make sure to upload your final 2-slide PPT. It carries **high weightage in the final evaluation** and is an important part of how your overall hackathon submission will be judged.

- Keep the final presentation to **2 slides only**.
- Use the slides to explain **what you built, what approach you took, and what you learned during the hackathon**.

- You can include things like **model choices, experiments tried, fine-tuning or prompting methods, frame/clip sampling strategy, temporal logic, optimizations, challenges, failures, and what finally worked.**
- Prefer **visuals over long text** - use simple graphs, comparison tables, timelines, example frames, before/after comparisons, or experiment progression.
- Show the **reasoning behind your choices.** If you changed models, prompts, sampling, or training strategy, explain briefly why.
- Highlight anything that makes your solution **faster, cheaper, more reliable, or better at reducing false alarms.**
- Keep it easy to scan. The judges should be able to understand your work in a few seconds without reading dense paragraphs.

Editable as often as you like, and it doesn't use benchmark runs.

Things that catch people out

1. Sending `"class_name": "normal"` gets rejected. Use `"events": []`.
2. Timestamps on Level-1 events get rejected. They must be `null`.
3. Expecting an omitted video to be cleared. It isn't. Re-submit that video to change it.
4. Many fragments for one event. Only the best-overlapping one can match.
5. Predicting anything on a normal Level-2/3 video. That video scores zero. False alarms are expensive by design.
6. Claiming the whole clip is anomalous. The 0.5 overlap gate puts this far below a real attempt.
7. Forgetting `runtime_metadata`. It is required on every video, and it's where your latency bonus comes from.

